

Aerial Teleoperation for Omnidirectional Tiltable-Quadrotors

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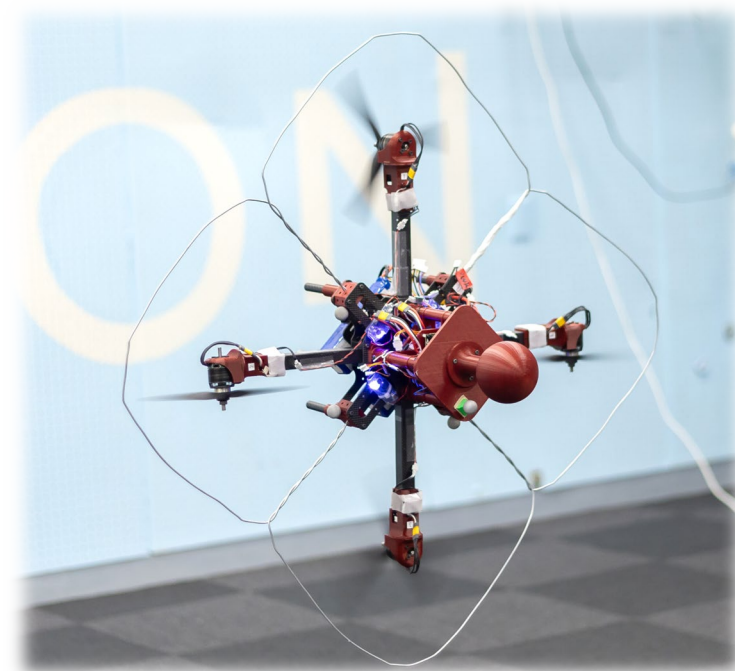


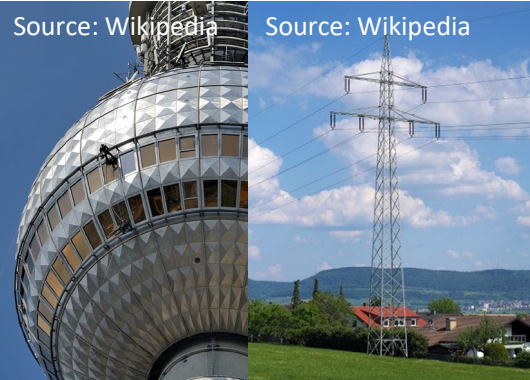
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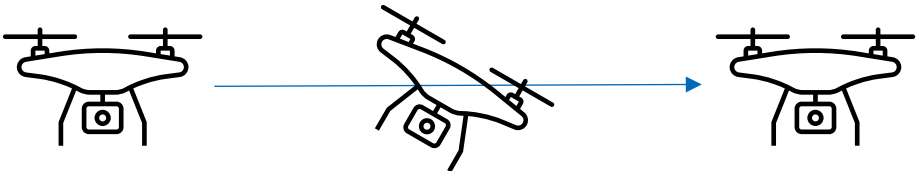
Introduction – Why tiltable-quadrotors?

Aerial Manipulation

Credit: DJI, MAVIC Air 2



Quadrotors: Underactuated



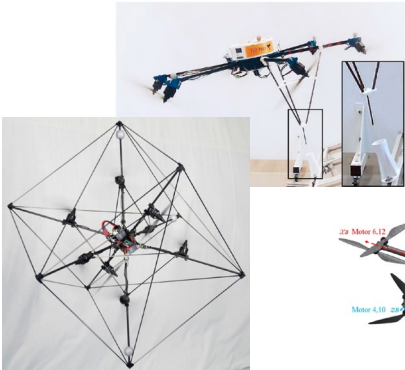
Independent Position & Attitude ❌

Reason: 4 (actuators) < 6 (DoF)

Full/Overactuation? Add actuators

Add Rotors

Ryll et al., IJRR, 2019

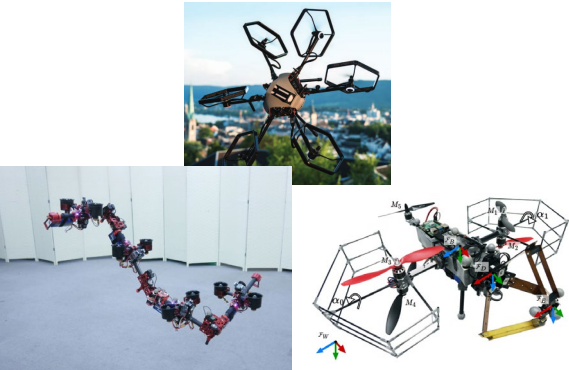


Brescianini et al., ICRA, 2016

Zhu et al., RA-L, 2024

Add Servos

Kamel et al., RA-M, 2018



Zhao et al., ICRA, 2018

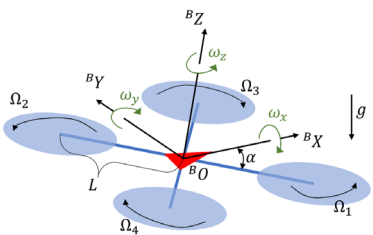
Cuniato et al., ICRA, 2023

Pros: Simplicity

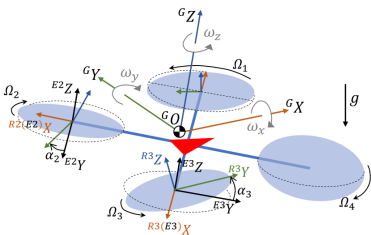
Cons: Inner Force; Efficiency

Pros: Efficiency

Cons: Complexity



Quadrotor



Tiltable-Quadrotor

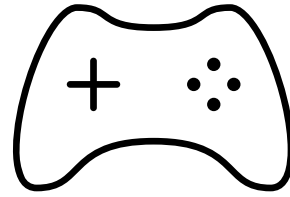


Ryll et al., T-CST, 2015

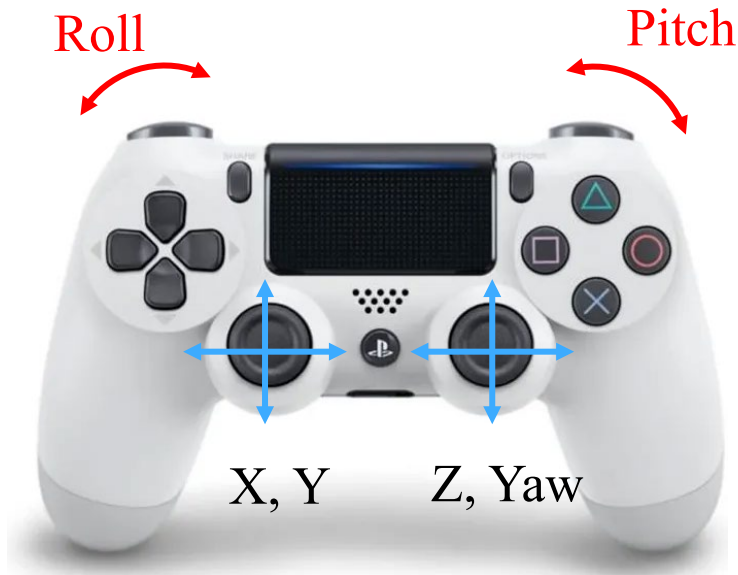
Effector-Centric NMPC of Tiltable Multirotors for Offset-Free **Omnidirectional** Aerial Manipulation

Pitch & Roll Full Rotation

Introduction – Omnidirectional Teleoperation



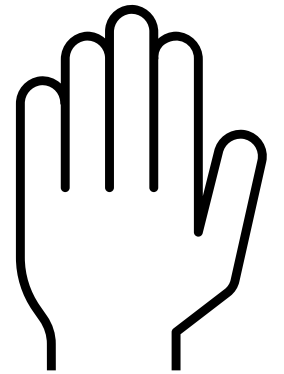
Demos



$$\mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R}^2 \cong \mathbb{R}^6$$

Not Equal !

$$SE(3) \cong \mathbb{R}^3 \times SO(3)$$

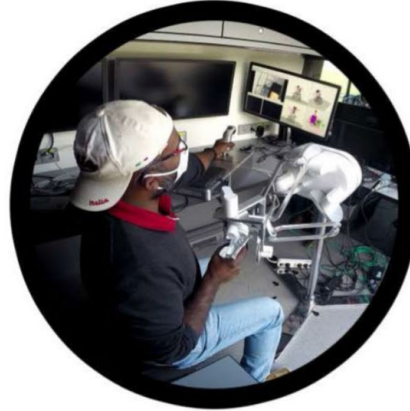


Introduction – Omnidirectional Teleoperation

Haptic Devices



J. Mellet, M. Tognon, et al., arXiv, 2024

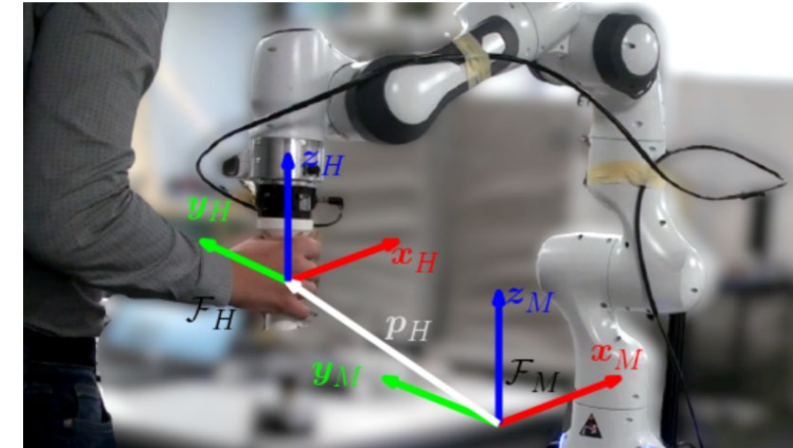


Lambda

J. Lee, et al., T-Field Robotics, 2024



Space Joystick

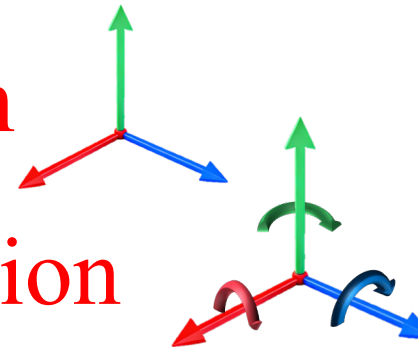


M. Allenspach, M. Tognon, et al., ICRA, 2022



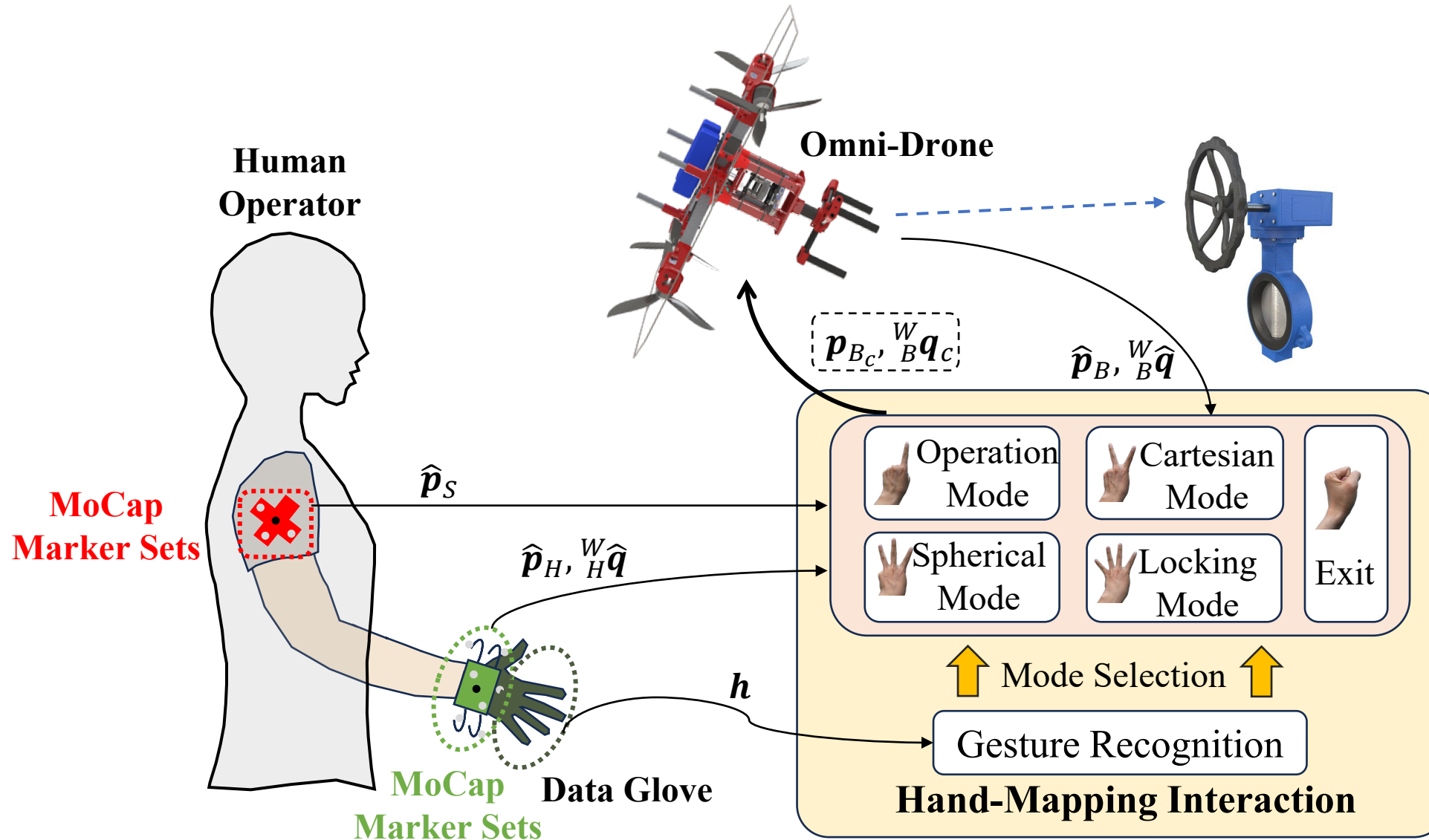
The fingers are constrained.

- ✓ Hand **position**
- ✓ Hand **orientation**
- Hand **gesture**



Finger freedom matters.

Omnidirectional Aerial Teleoperation





Six-DoF Hand-Based Teleoperation for Omnidirectional Aerial Robots

Jinjie Li^{†*}, Jiaxuan Li^{2†}, Kotaro Kaneko^{1,3}, Haokun Liu¹, Liming Shu², and Moju Zhao¹

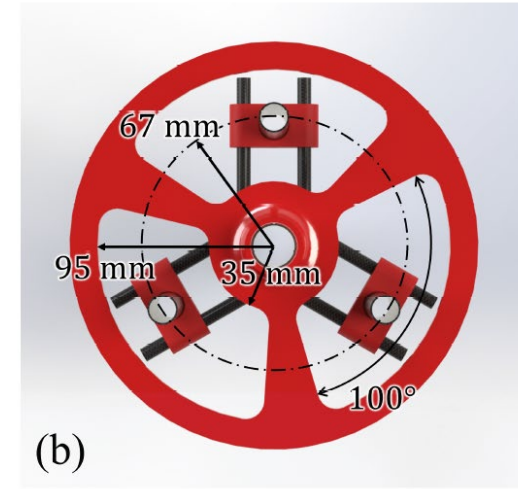
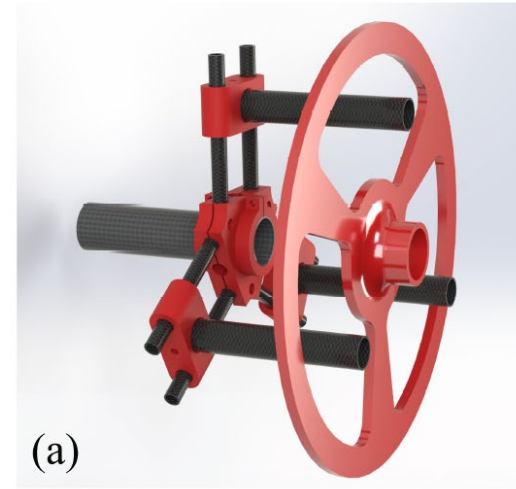
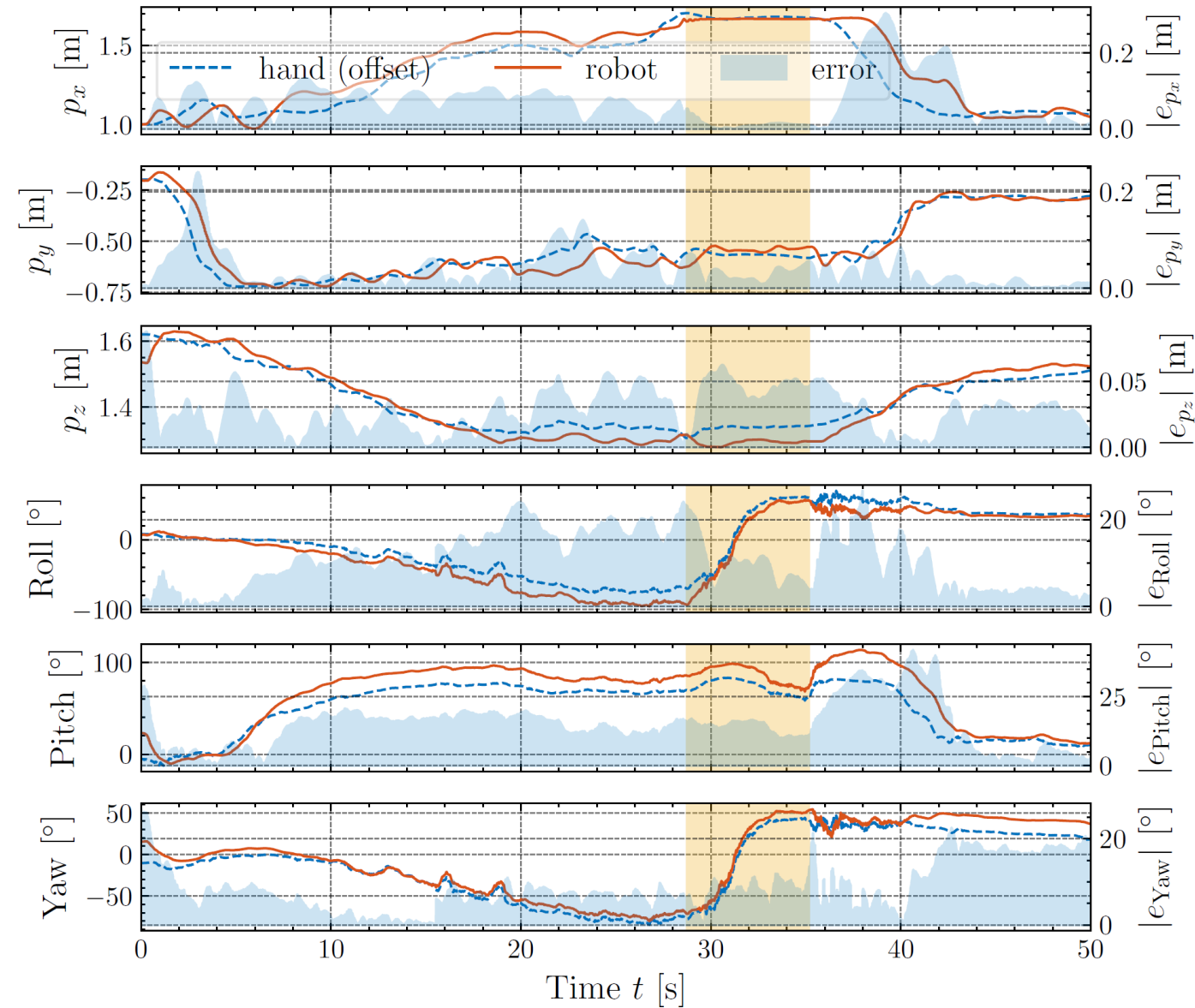
[†]Equal Contributions

¹DRAGON Lab, Department of Mechanical Engineering, The University of Tokyo

²Intelligent Equipment and Medical Device Lab, Department of Mechanical Engineering, Dalian University of Technology

³HNL Lab, Department of Mechanical Engineering, The University of Tokyo

Result



- ❑ Control is stable but has offset
- ❑ RMSEs (all positional axes) are over 3 cm, $>$ maximum tolerance
- ❑ Human operator can deal with this offset and ensure the safety

Conclusion

Six-DoF Hand-Based Teleoperation for Omnidirectional Aerial Robots

IROS 2025, Hangzhou

Key Takeaways

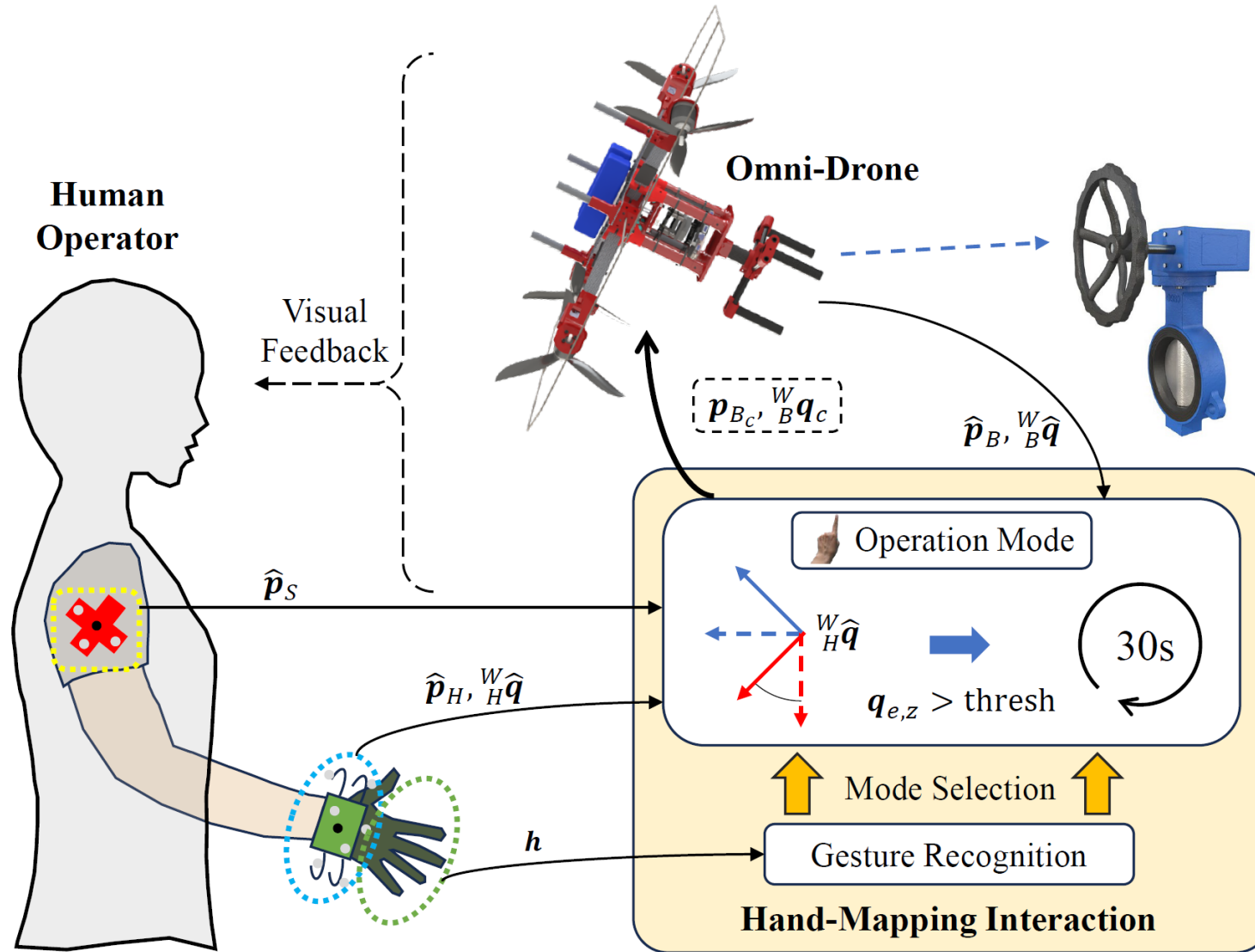
- To our best knowledge, this is **the first time** the rotational dexterity of human wrist → an omnidirectional aerial robot.
- Precise operation: Locking mode matters!
- Long-range moving: Spherical mode → natural, and Cartesian mode → moving along regular geometries like squares.

Future Directions

- Shared control: rotations that exceed the natural flexibility
- Force and torque support
- Fully onboard flight; video streaming

Workshop Friday 106 ~12:30

Shared-Control Teleoperation





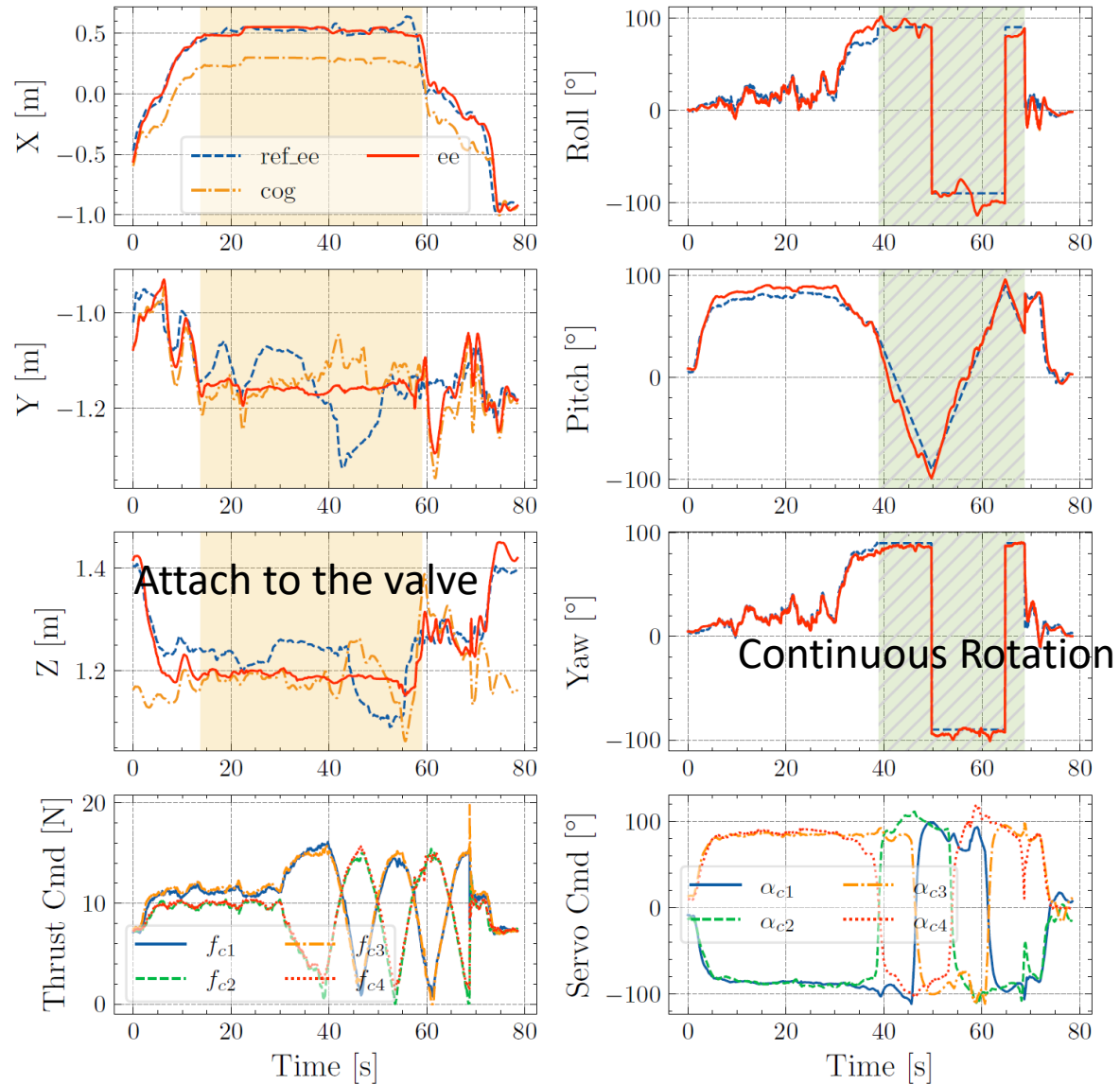
IROS 2025 Workshop: Advancements in Aerial Physical Interaction

Shared-Control Teleoperation of Omnidirectional Aerial Robots for Continuous Valve Rotation

Jinjie Li, Yicheng Chen, Haokun Liu, and Moju Zhao

DRAGON Lab, Department of Mechanical Engineering, The University of Tokyo

Shared-Control Teleoperation



Constant force and torque exist, requiring compliance behaviors

Effector-Centric NMPC of Tiltable Multirotors for Offset-Free Omnidirectional **Aerial Manipulation**

Pushing a whiteboard

Thank you for listening



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Research Interests: Robotics, Model Predictive Control, Aerial Manipulation



About Me

Greetings! I am currently a second-year PhD student in the [DRAGON Lab](#), [Department of Mechanical Engineering](#), at [The University of Tokyo](#), under the supervision of [Junior Associate Prof. Moju ZHAO](#).

My research interest lies in optimization-based control with applications in aerial manipulation, aiming to make aerial robots function as flying hands rather than just eyes.

I obtained an M.Sc. in Control Science and Engineering (2023) and a B.Eng. in Automation (2020) from [Beihang University](#). Please refer to my [CV](#) for more information.

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